

Personal Portfolio Website - Anusha Tyarla

Project Overview

Description

A responsive personal portfolio website showcasing skills, projects, and contact information. Built with HTML5, CSS3, and JavaScript following modern web design principles.

Goals and Objectives

- Create a professional online presence to showcase technical skills and projects
- Demonstrate proficiency in HTML5, CSS3, and responsive web design
- Implement accessible and semantic markup following web standards
- Build a fully functional contact form with client-side validation
- Ensure cross-browser compatibility and mobile responsiveness
- Apply modern design trends including glassmorphism and smooth animations

Key Features

- Fully responsive design (mobile, tablet, desktop)
 - Semantic HTML5 structure
 - CSS Grid and Flexbox layouts
 - Interactive navigation with mobile hamburger menu
 - Animated background with floating shapes
 - Contact form with validation
 - Smooth scrolling and hover effects
 - Social media integration
-

Setup Instructions

Step-by-Step Installation and Configuration Guide

Prerequisites

- Modern web browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- Text editor (VS Code recommended)
- Git (optional, for version control)

Step 1: Download/Clone the Project

```
bash

# Using Git
git clone https://github.com/TyarlaAnusha/portfolio.git
cd portfolio
# Or download ZIP and extract
```

Step 2: Verify File Structure

```
portfolio/
├── index.html
├── css/
│   └── style.css
├── js/
│   └── navigation.js
├── asset/
│   └── hero.png
├── README.md
└── .gitignore
```

Step 3: Configuration

- Update personal information in `index.html`
- Replace `asset/hero.png` with your photo
- Modify colors in `css/style.css` (CSS variables at top)
- Update contact email in form action attribute

Step 4: Open in Browser

bash

Option 1: Direct open

Double-click index.html

Option 2: Using VS Code Live Server

Install "Live Server" extension

Right-click index.html → "Open with Live Server"

Option 3: Using Python

python -m http.server 8000

Visit http://localhost:8000

Code Structure

File Hierarchy

```
portfolio/
|
├── index.html      # Main HTML document
|   ├── <header>    # Navigation bar
|   ├── <main>      # Main content sections
|       ├── Hero section
|       ├── About section
|       ├── Experience timeline
|       ├── Projects grid
|       ├── Skills categories
|       ├── Certificates
|       └── Contact form
|   └── <footer>    # Social links & copyright
|
├── css/
|   ├── style.css   # Stylesheet (1100+ lines)
|   ├── CSS Variables # Theme colors
|   ├── Reset styles # Global resets
|   ├── Animations   # Keyframes & transitions
|   ├── Components   # Navigation, cards, buttons
|   ├── Sections     # Hero, about, projects, etc.
|   └── Media queries # Responsive breakpoints
|
└── js/
```

```
| └─ navigation.js    # JavaScript functionality
|   └─ Mobile menu toggle
|   └─ Smooth scrolling
|   └─ Form validation
|   └─ Scroll effects
|
└─ asset/
   └─ hero.png        # Profile image
   └─
└─ README.md          # Documentation
└─ .gitignore          # Git exclusions
```

Component Architecture

Navigation Component

```
<nav>
├─ Logo (text-based with gradient)
├─ Navigation links (6 items)
└─ Mobile burger menu (JavaScript-driven)
```

Section Pattern

```
<section id="[name]" class="section">
  <div class="container">
    <h2 class="section-title">Title</h2>
    <div class="[content-grid]">
      <!-- Cards/Content -->
    </div>
  </div>
</section>
```

Technical Details

Architecture Overview

1. HTML5 Semantic Structure

- Uses semantic elements: `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, `<footer>`
- Proper heading hierarchy (h1 → h2 → h3)

- ARIA labels for accessibility
- Alt attributes for all images

2. CSS Architecture

CSS Custom Properties (Variables)

```
css

:root {
  --primary: #6366f1; /* Indigo blue */
  --secondary: #ec4899; /* Pink */
  --accent: #f59e0b; /* Amber */
  --dark: #0f172a; /* Dark background */
  --light: #f8fafc; /* Light text */
}
```

Layout System

- **Flexbox:** Used for navigation, hero content, skill items, buttons
- **CSS Grid:** Used for about section (1fr 2fr), projects (auto-fit minmax), skills (auto-fit 300px)
- **Positioning:** Fixed navigation, absolute floating shapes, relative containers

Responsive Strategy

- Mobile-first approach
- Single breakpoint at 768px
- Fluid typography using rem units
- Flexible images (max-width: 100%)

3. Animation System

Keyframe Animations

```
css
```

```

@keyframes float {
  /* 6 shapes float from bottom to top */
  /* Duration: 20s, infinite loop */
}

@keyframes slideUp {
  /* Elements fade in while sliding up */
  /* Staggered delays: 0.2s, 0.4s, 0.6s, 0.8s */
}

```

CSS Transitions

- Navigation links: 0.3s ease
- Cards hover: transform + box-shadow
- Buttons: transform + shadow on hover
- Form inputs: border color + background

4. JavaScript Functionality

Mobile Navigation Logic

```

javascript

// Toggle menu open/close
burger.click → navLinks.toggle('active')

// Burger animation
line1: rotate(-45deg) + translate
line2: opacity(0)
line3: rotate(45deg) + translate

```

Form Validation Algorithm

```

javascript

1. Check all fields not empty (trim whitespace)
2. Validate email format using regex:
  /^[^s@]+@[^s@]+\.[^s@]+$ /
3. If valid → allow submission
4. If invalid → preventDefault() + show alert

```

Smooth Scroll Implementation

```
javascript
```

```
// Intercept anchor clicks
```

```
anchor.click → preventDefault()
```

```
target.scrollIntoView({ behavior: 'smooth' })
```

5. Data Structures Used

Project Data (Implicit structure in HTML)

```
javascript
```

```
{  
  icon: "emoji/icon",  
  title: "Project Name",  
  date: "Year range",  
  description: "Brief description",  
  techStack: ["Tech1", "Tech2", ...]  
}
```

Skills Categories (6 categories)

```
javascript
```

```
{  
  category: "Category Name",  
  icon: "FontAwesome class",  
  skills: ["Skill1", "Skill2", ...]  
}
```

Performance Optimizations

- Minimal external dependencies (only Font Awesome + Google Fonts)
- CSS animations use `transform` and `opacity` (GPU-accelerated)
- Backdrop-filter for glassmorphism effect
- No JavaScript libraries (vanilla JS only)
- Images optimized for web

Accessibility Features

- Semantic HTML for screen readers
- ARIA labels on icon-only links

- Keyboard navigation support
 - Focus states on interactive elements
 - Sufficient color contrast (WCAG AA)
 - Alt text for all images
-

Testing Evidence

Browser Compatibility Testing

Test Results:

- Navigation: All links work, smooth scroll functional
- Animations: Floating shapes render correctly
- Forms: Validation works, email format checked
- Responsiveness: Adapts to all screen sizes

Device Testing

Mobile Devices:

Device	Screen Size	Status	Issues
Galaxy S21	360x800	Pass	Burger menu works
iPad Pro	1024x1366	Pass	Tablet layout optimal

Desktop Resolutions:

- 1920x1080 (Full HD): Pass
- 2560x1440 (2K): Pass
- 1366x768 (Laptop): Pass

Responsiveness Test Cases

Test Case 1: Navigation on Mobile

Steps:

1. Open site on mobile (< 768px)
2. Click burger menu
3. Menu slides in from right
4. Click any link
5. Menu closes, scrolls to section

Expected: Menu toggles, links work, smooth scroll

Actual: Passed

Test Case 2: Form Validation

Test 2a: Empty Fields

Input: Submit with empty fields

Expected: Alert "Please fill in all fields"

Actual: Passed

Test 2b: Invalid Email

Input: name@invalid

Expected: Alert "Please enter a valid email"

Actual: Passed

Test 2c: Valid Submission

Input: All fields filled correctly

Expected: Opens email client with pre-filled data

Actual: Passed

Test Case 3: Hover Effects

Element: Project cards

Action: Mouse hover

Expected: Card lifts up 10px, border color changes

Actual: Passed - Transform and shadow applied

Element: Navigation links

Action: Mouse hover

Expected: Underline animates from left to right

Actual: Passed - Width transition works

Technologies Used

- **HTML5** - Semantic markup
 - **CSS3** - Modern styling with Grid, Flexbox, animations
 - **JavaScript** - Vanilla JS for interactivity
 - **Font Awesome 6.0** - Icon library
 - **Google Fonts (Inter)** - Typography
 - **Git** - Version control
-

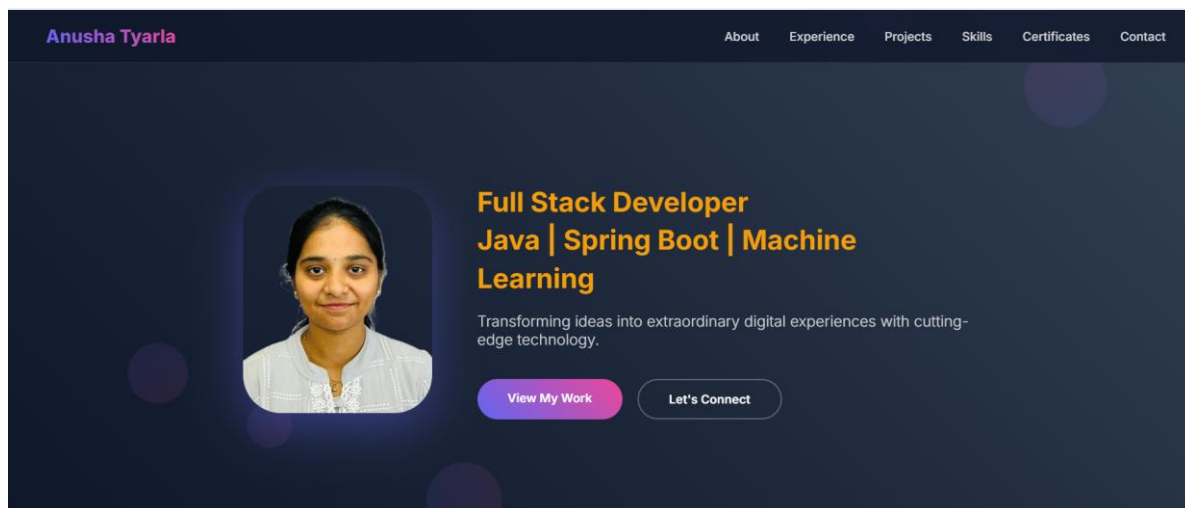
Contact

Anusha Tyarla

- ♦ Email: tyarla.anusha@gmail.com
 - ♦ LinkedIn: [linkedin.com/in/tyarla-anusha-77a271227](https://www.linkedin.com/in/tyarla-anusha-77a271227)
 - ♦ GitHub: github.com/TyarlaAnusha
-

Visual Documentation

Homepage - Hero Section



About Section Layout

Anusha Tyarla

AboutExperienceProjectsSkillsCertificatesContact

About Me

Education

Malla Reddy University

B.Tech in Computer Science Engineering

2020 – 2024 | CGPA: 8.75/10

Hyderabad, India

Passionate Full Stack Developer with hands-on experience in building scalable web applications.

My journey spans across full-stack development, AI/ML innovations, and competitive programming.

I have built systems like Hospital Management System (reduced 70% manual work) and ML based apps achieving 92% accuracy.

Experience Timeline (Desktop)

Anusha Tyarla

AboutExperienceProjectsSkillsCertificatesContact

Experience Journey

Java Full Stack Trainee

Codegnan IT Solutions, Hyderabad | Aug 2024 – Mar 2025

→ Built and deployed Hospital Management Web App reducing manual record management by 70%.

→ Optimized backend performance by 30% using caching and efficient SQL queries.

→ Delivered sprint modules ahead of schedule.

→ Achieved A+ grade for technical performance.

Projects Grid

Anusha Tyarla

AboutExperienceProjectsSkillsCertificatesContact

Featured Projects

Hospital Management System

2024 - 2025

Spring Boot-based HMS with REST APIs using MVC architecture.

Spring BootMySQLRESTful APISMavenPostman

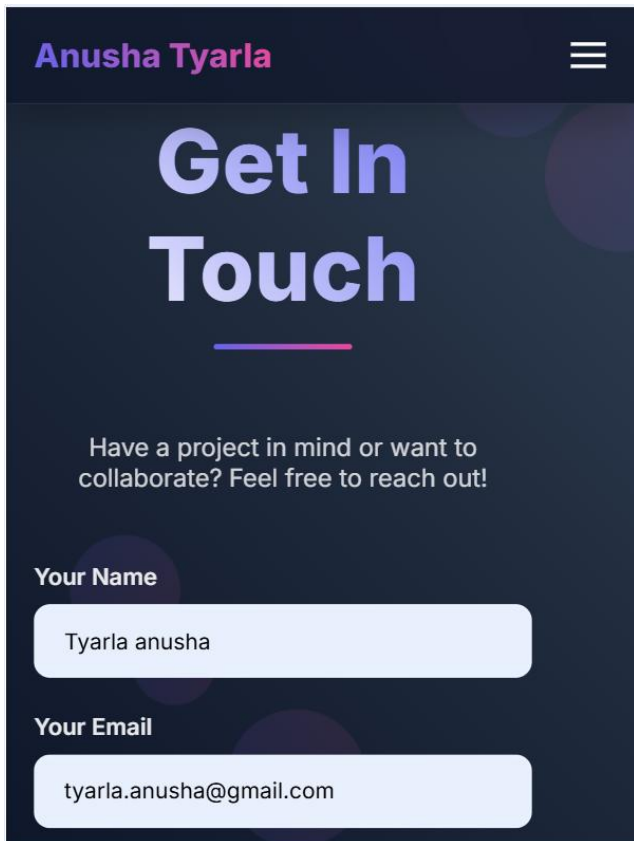
Fertilizer Management System

2023 - 2024

Angular + MySQL system providing fertilizer recommendations.

AngularMySQLTypeScriptHTML/CSS

Mobile View (< 768px)



The image shows a mobile view of a contact form. The background is dark blue with faint, overlapping circles in shades of purple and blue. At the top left, the name 'Anusha Tyarla' is written in a pinkish-purple font. To the right of the name is a white hamburger menu icon. Below the name, the text 'Get In Touch' is displayed in a large, bold, white sans-serif font. A thin horizontal line, colored with a gradient from blue to pink, is positioned below the text. Underneath the line, the text 'Have a project in mind or want to collaborate? Feel free to reach out!' is written in a smaller, white sans-serif font. Further down, the label 'Your Name' is shown in a white sans-serif font. Below this label is a light blue rounded rectangular input field containing the text 'Tyarla anusha'. Below the input field, the label 'Your Email' is shown in a white sans-serif font. Below this label is another light blue rounded rectangular input field containing the text 'tyarla.anusha@gmail.com'.

Anusha Tyarla

Get In Touch

Have a project in mind or want to collaborate? Feel free to reach out!

Your Name

Tyarla anusha

Your Email

tyarla.anusha@gmail.com

